

DATA SOURCE DETAILS

Early Chronic-Disease Risk Prediction — Diabetes

1. Dataset Overview

The project uses the Pima Indians Diabetes Dataset for early risk stratification of Type-2 diabetes. The report identifies this as a public dataset containing 768 patient records, 8 clinical input features, and a binary outcome.

Item	Details
Dataset Name	Pima Indians Diabetes Dataset
Primary Source	UCI Machine Learning Repository
Mirror Mentioned in Report	Kaggle mirror
Number of Records	768
Input Features	8 clinical features
Target Variable	Outcome
Problem Type	Supervised binary classification
Application	Early Type-2 diabetes risk prediction
Population	Pima Indian women, age ≥ 21

2. Dataset Features

No.	Feature	Description
1	Pregnancies	Number of pregnancies
2	Glucose	Plasma glucose concentration
3	Blood Pressure	Diastolic blood pressure
4	Skin Thickness	Triceps skinfold thickness
5	Insulin	2-hour serum insulin
6	BMI	Body Mass Index
7	Diabetes Pedigree Function	Diabetes pedigree function
8	Age	Patient age
9	Outcome	Binary diabetes outcome/label: 0 or 1

3. Target Variable

The target variable is Outcome, which represents the binary classification of the patient. In the dataset, 0 represents the negative class and 1 represents the positive/diabetes class.

4. Data Preprocessing Required by the Report

- Zero-valued entries in Glucose, Blood Pressure, Skin Thickness, Insulin and BMI are treated as missing values.
- Missing values are imputed rather than dropping records, preserving the dataset size.
- The report specifies class-wise median imputation.
- Features are standardised to zero mean and unit variance before MLP and Gaussian Naive Bayes processing.
- Class imbalance is addressed using class-weighted loss for the MLP and configurable class-balanced priors for Naive Bayes.
- The evaluation design uses an 80/20 stratified hold-out and stratified 5-fold cross-validation within the training data.

5. Dataset Role in the Proposed Framework

The dataset is used to evaluate an adaptive multi-paradigm learning framework that combines a Multilayer Perceptron (MLP), a Genetic Algorithm (GA) optimiser, and a Naive Bayes (NB) probabilistic baseline. The MLP provides a discriminative learning model, the GA searches for effective initialisation/hyperparameter settings, and Naive Bayes provides a probabilistic complementary model. Their outputs are combined using reliability-weighted probability fusion.

6. Data Source Reference

The report lists the following source in its references: UCI Machine Learning Repository, "Pima Indians Diabetes Dataset." The report also describes the available data as the Pima Indians Diabetes Dataset (UCI/Kaggle mirror).

7. Source-Based Summary

The selected dataset is appropriate for the stated problem because it contains clinical and demographic variables together with a binary diabetes outcome. The project uses these records as the common input for MLP, GA-assisted MLP, Naive Bayes, and the final fusion model.

8. Note on Scope

All details in this document are based on the uploaded project report. The report identifies the UCI Machine Learning Repository as the dataset source and also mentions a Kaggle mirror; it does not provide a more specific dataset URL than the UCI repository address in its reference section.